

Componente Curricular: Física - Dinâmica

Professor(a): Roxana Lúcia

Série: 2ª

Data: Abril / Maio

Conteúdo: Teorema da Energia Cinética (TEC) - Resolução

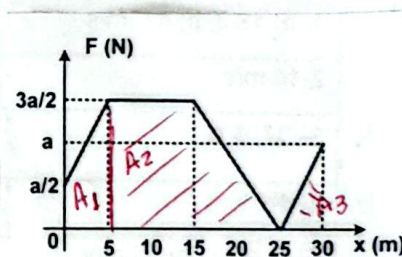
1. $m = 4 \text{ kg}$ $d = 3 \text{ m}$ a) $\bar{E}_R = \bar{E}_F + \bar{E}_{Fat}$
 $F = 15,2 \text{ N}$ $v_N = 5 \text{ m/s}$ $\bar{E}_R = F \cdot d + (-F_{at} \cdot d)$
 $F_{at} = 9,2 \text{ N}$ $\bar{E}_R = 15,2 \cdot 3 + (-9,2 \cdot 3)$
 $\bar{E}_R = 45,6 - 27,6 \Rightarrow \bar{E}_R = 18 \text{ J}$

b) $\bar{E}_R = \Delta E_c$
 $\bar{E}_R = \frac{mv^2}{2} - \frac{mv_0^2}{2}$
 $18 = \frac{4 \cdot v^2}{2} - \frac{4 \cdot 5^2}{2}$
 $18 = 2v^2 - 50 \Rightarrow v^2 = 34 \Rightarrow v = \sqrt{34} \text{ m/s}$

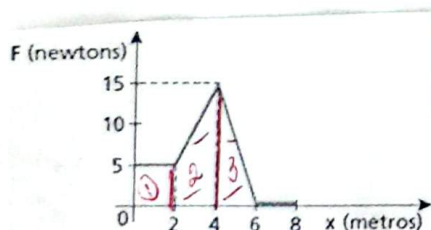
2. $m = 16,8 \text{ kg}$
 $v_0 = 0$
 $v = ?$
 $\bar{E}_R = \Delta E_c \Rightarrow A = \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow \left(\frac{10+4}{2}\right) \cdot 120 = \frac{16,8 \cdot v^2}{2} \Rightarrow 1680 = 16,8 v^2 \Rightarrow v^2 = 100 \Rightarrow v = 10 \text{ m/s}$

3. $m = 6 \text{ kg}$
 $v_0 = 0$
 $v = 5 \text{ m/s}$
 $\bar{E}_R = \Delta E_c \Rightarrow A = \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow \left(\frac{3+1}{2}\right) \cdot F_0 = \frac{6 \cdot 5^2}{2} \Rightarrow 4 F_0 = 150 \Rightarrow F_0 = 37,5 \text{ N}$

4. $\bar{E}_R = A \Rightarrow 120 = \left(\frac{\frac{3a}{2} + \frac{a}{2}}{2}\right) \cdot 5 + \left(\frac{20+10}{2}\right) \cdot \frac{3a}{2} + \frac{5 \cdot a}{2}$
 $120 = 5a + 22,5a + 25a \Rightarrow 120 = 30a \Rightarrow a = 4 \text{ N}$



5.



a) $\bar{E}_{0,6} = A_1 + A_2 + A_3 \Rightarrow \bar{E}_{0,6} = 5 \cdot 2 + \frac{(5+15) \cdot 2}{2} + \frac{2 \cdot 15}{2} \Rightarrow \bar{E}_{0,6} = 10 + 20 + 15$
 $\bar{E}_{0,6} = 45 \text{ J}$

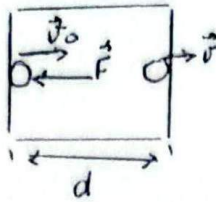
b) 0 a 6m

$\bar{E}_R = \Delta E_c \Rightarrow \bar{E}_R = \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow 45 = \frac{0,9 \cdot v^2}{2} \Rightarrow v^2 = 100 \Rightarrow v = 10 \text{ m/s}$
 $x = 6 \text{ m}$

De 6m a 8m $\Rightarrow F = 0 (a = 0) \Rightarrow v = \text{cte}$

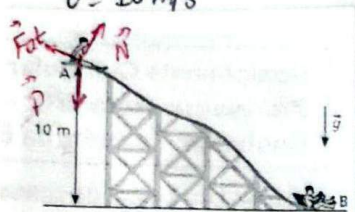
$\therefore v_{6m} = v_{8m} \Rightarrow v_{8m} = 10 \text{ m/s}$

6. $m = 10g = 0,01 \text{ kg}$
 $v = 100 \text{ m/s}$
 $v_0 = 600 \text{ m/s}$
 $e = d = 5 \text{ cm} = 0,05 \text{ m}$



$$\begin{aligned} \bar{E}_R &= \Delta E_c \\ -F \cdot d &= \frac{mv^2}{2} - \frac{mv_0^2}{2} \\ -F \cdot 0,05 &= \frac{0,01 \cdot 100^2}{2} - \frac{0,01 \cdot 600^2}{2} \\ -F \cdot 0,05 &= 50 - 1800 \Rightarrow 0,05 F_0 = 1750 \Rightarrow F_0 = 35000 \text{ N} \end{aligned}$$

7. $m = 40 \text{ kg}$
 $v_0 = 0$
 $v = 10 \text{ m/s}$

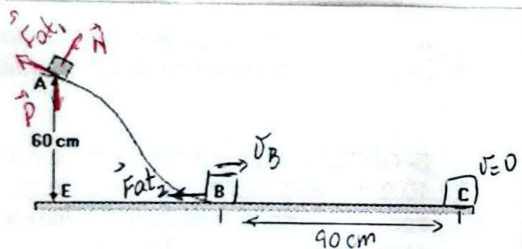


$$\begin{aligned} \bar{E}_R &= \bar{E}_P + \bar{E}_N + \bar{E}_{Fat} \\ \bar{E}_R &= P \cdot h + \bar{E}_{Fat} \\ \bar{E}_R &= \Delta E_c \Rightarrow \bar{E}_R = \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow \\ \Rightarrow \bar{E}_R &= \frac{40 \cdot 10^2}{2} \Rightarrow \boxed{\bar{E}_R = 2000 \text{ J}} \end{aligned}$$

$$\begin{aligned} \therefore 2000 &= 400 \cdot 10 + \bar{E}_{Fat} \\ 2000 &= 4000 + \bar{E}_{Fat} \Rightarrow \boxed{\bar{E}_{Fat} = -2000 \text{ J}} \end{aligned}$$

8. $m = 0,2 \text{ kg}$

9. $h = 0,6 \text{ m}$
 $v_0 = 2 \text{ m/s}$
 $v_B = 3 \text{ m/s}$

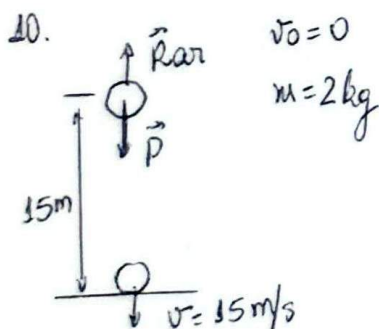


$$\bar{E}_R = \Delta E_c \Rightarrow \bar{E}_P + \bar{E}_N + \bar{E}_{Fat} = \frac{mv^2}{2} - \frac{mv_0^2}{2}$$

$$\begin{aligned} P \cdot h + \bar{E}_{Fat}_1 &= \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow 2 \cdot 0,6 + \bar{E}_{Fat}_1 = \frac{0,2 \cdot 3^2}{2} - \frac{0,2 \cdot 2^2}{2} \Rightarrow \\ \Rightarrow 42 + \bar{E}_{Fat}_1 &= 0,9 - 0,4 \Rightarrow \boxed{\bar{E}_{Fat}_1 = -0,7 \text{ J}} \end{aligned}$$

$$\bar{E}_R = \Delta E_c$$

$$\begin{aligned} \bar{E}_P + \bar{E}_N + \bar{E}_{Fat}_2 &= \frac{mv^2}{2} - \frac{mv_0^2}{2} \\ -F_{at} \cdot d &= \frac{mv^2}{2} - \frac{mv_0^2}{2} \Rightarrow -\mu m g d = \frac{mv^2}{2} - \frac{mv_0^2}{2} \\ -\mu \cdot 10 \cdot 0,9 &= -\frac{3^2}{2} \Rightarrow 9\mu = 4,5 \Rightarrow \mu = 0,5 \end{aligned}$$



$$\bar{E}_R = \Delta E_c \Rightarrow \bar{E}_P + \bar{E}_{Rar} = \frac{mv^2}{2} - \frac{mv_0^2}{2}$$

$$P \cdot h + \bar{E}_{Rar} = \frac{mv^2}{2}$$

$$20 \cdot 15 + \bar{E}_{Rar} = \frac{2 \cdot 15^2}{2} \Rightarrow 300 + \bar{E}_{Rar} = 225$$

$$\boxed{\bar{E}_{Rar} = -75 \text{ J}}$$